

GenAI Hackathon

Trust-Aware Memory Intelligence System

Background

Modern AI systems can generate answers, retrieve documents, and interact with users. However, most systems struggle with a harder problem: how should an AI system decide what to remember, what to update, what to distrust, and what to forget over time?

In this challenge, participants will build a Trust-Aware Memory Intelligence System capable of ingesting noisy, conflicting, and evolving information streams and transforming them into a trusted, provenance-aware memory.

This is not a chatbot challenge and not a traditional RAG application.

The focus is on designing a memory system that continuously reasons about changing information and evolves its beliefs as new evidence arrives.

Problem Statement

Build a multi-agent memory intelligence system that ingests a stream of noisy, conflicting claims and converts them into a structured memory that improves over time.

The system must continuously determine:

- What to store
- What to revise
- What to downgrade
- What to merge
- What to forget

Every memory entry should maintain:

- Confidence score
- Source information
- Timestamp
- Current status
- Provenance history

The system should be able to explain why it believes a fact at any point in time.

Objective

Design an agentic pipeline that transforms unreliable, time-ordered inputs into a trusted memory store.

A strong solution should demonstrate how a single fact can:

- Be accepted
- Be revised
- Be downgraded
- Be rejected
- Be forgotten

Core Capabilities to Build

1. Required System Components

- Claim Extraction Agent
- Verification Agent
- Contradiction Detection Agent
- Memory Curator Agent

2. Dataset Scenario

Participants will receive a synthetic "Evolving Truth" dataset containing repeated entities, conflicting updates, duplicate claims, missing timestamps, adversarial statements, noisy information, and memory overflow scenarios.

Example Stream:

- T1: Startup A raised \$5M in 2021
- T2: Startup A raised \$8M in 2021
- T3: Startup A raised \$8M in 2022
- T4: Startup A failed to raise funding

3. Expected Deliverables

- Memory Store
- Change Log
- Demonstration Flow

4. Constraints

- Stateful memory is mandatory
- Pure prompting is insufficient
- Explicit memory logic is required
- Explainability matters

5. Recommended Technology Stack

LLMs: OpenAI, Anthropic, Open-source Models

Frameworks: LangGraph, LangChain

Storage: Neo4j, Redis, Valkey, PostgreSQL, JSON

Backend: Python

6. Edge Cases

- Same fact with different wording
- Repeated contradictions
- High-confidence incorrect claims
- Missing timestamps
- Equal-confidence conflicts
- Adversarial inputs
- Memory overflow situations

Bonus Opportunities

Teams can strengthen their solution with supporting enhancements such as richer provenance visualization, stronger contradiction handling, memory compression strategies, automated scoring, and demo-ready evaluation flows.

Expected Outcome

The final solution should demonstrate how agentic AI can transform unreliable, evolving inputs into a structured, trustworthy, and explainable memory system that tracks belief changes over time.

The goal is not to build the smartest chatbot.

The goal is to build a memory system that can answer:

“Why do I believe this fact right now, and how has that belief changed over time?”

The best solutions will demonstrate trust, provenance, memory evolution, and explainable reasoning under uncertainty.